# Thirukkural AI – Domain-Specific Tamil Knowledge Model

## Project Statement

Classical Tamil literature, prominently திருக்குறள் (Thirukkural), represents the foundational corpus of human ethics, governance, social justice, and universal moral philosophy. Composed of **1,330 couplets (குறட்பாக்கள்)** categorized across **133 chapters (அதிகாரங்கள்)**, **9 sub-sections (இயல்கள்)**, and **3 primary divisions (அறத்துப்பால், பொருட்பால், காமத்துப்பால்)**, it stands as an unexcelled guide to righteous living.

Despite its profound significance, access to Thirukkural remains constrained for modern learners, competitive exam aspirants (UPSC / TNPSC), and the global diaspora due to the poetic complexity of classical Tamil (செந்தமிழ்), fragmented traditional print commentaries, and the absence of dependable AI-based interpretation platforms.

Most existing digital tools are static text repositories or unconstrained generic LLMs prone to severe hallucinations, generating fictional verses or incorrect moral attributions. There is an urgent need for a **Tamil-first, explainable, and academically grounded AI system** that makes Thirukkural accessible, interactive, and trustworthy in simple modern Tamil.

To address this gap, we propose **Thirukkural AI (திருக்குறள் நிபுணர்)**, an intelligent domain-specific Tamil Language and Retrieval Model engineered for zero-hallucination semantic search, comparative multi-scholar commentary synthesis, and multi-modal image intelligence.

## Proposed Solution

**Thirukkural AI** is an LLM-powered conversational and research platform designed to interact with the complete Thirukkural corpus and provide insightful explanations in **எளிய இன்றைய தமிழ் (Simple Modern Tamil)** and English without compromising classical literary authenticity.

Rather than relying on ungrounded generation, the platform integrates a **Retrieval-Augmented Generation (RAG)** framework with local vector embeddings and deterministic scholarly knowledge bases. Authoritative commentaries from **மு. வரதராசனார் (Mu. Va.)**, **சாலமன் பாப்பையா (Solomon Pappaiah)**, **மு. கருணாநிதி (Kalaignar)**, and **Rev. G.U. Pope** serve as the verified knowledge core, ensuring **உரை அடிப்படையிலான நம்பகமான மற்றும் துல்லியமான விளக்கங்கள் (Rigorous, Verified Commentary-Grounded Explanations)**.

When a user submits a natural language question, an Athigaram name/number (e.g., *அதிகாரம் 13*), or an image of a verse, the system retrieves the exact couplets and multi-scholar commentaries. The retrieved context is provided to the Tamil AI engine to synthesize structured, hallucination-free explanations.

**The system delivers:**

- Deterministic Athigaram & Verse Resolution (retrieving all 10 Kurals per chapter with zero hallucination).
- Comparative Multi-Scholar Commentary (மு.வ., சாலமன் பாப்பையா, கலைஞர் உரை, ஆங்கில மொழிபெயர்ப்பு).
- Paal, Iyal, and Athigaram hierarchical classification.
- Real-time English-to-Tamil phonetic transliteration typing.
- Vision OCR understanding for manuscript and exam question image inputs.

**User Input:** Tamil Text / English Transliteration / Manuscript Image

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**Preprocessing Layer:** Unicode Normalization / OCR / Intent Router

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**Retrieval Module (RAG):** Hybrid Vector Search & Athigaram Deterministic Engine

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**Grounded Tamil LLM: Context-Bound Zero-Hallucination Decoder**

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**Structured Output:** Metrical Verse + 3 Scholar Commentaries + Moral Meaning

## Target Beneficiaries

Thirukkural AI will empower a broad spectrum of users, including:

- **Civil Services (UPSC) Aspirants:** Tamil Literature optional & Ethics (GS-IV) essay citations.
- **Tamil Nadu Group 1, Group 2, Group 3, and Group 4 Candidates:** Compulsory Tamil eligibility and descriptive questions.
- **Tamil Scholars, Academicians & Literary Researchers:** Cross-commentary comparative studies.
- **School & University Students:** Simplified, relatable ethical education.
- **Global Tamil Diaspora & Heritage Learners:** Bilingual Tamil ↔ English access.

## Core Functionality & Model Capabilities

- **Domain-Specific Tamil RAG Architecture:** Grounded strictly on 1,330 Kurals and verified commentaries.
- **Multi-Scholar Comparative Insight:** Side-by-side synthesis of Mu.Va, Solomon Pappaiah, and Kalaignar Karunanidhi.
- **Real-Time Phonetic Transliteration:** Instant Tanglish-to-Tamil conversion on spacebar press.
- **Vision OCR Input:** Image-based query grounding for printed poems, books, or handwritten text.
- **3D Cultural Heritage Experience:** Interactive 3D Thiruvalluvar model with dynamic lighting.
- **Offline PWA & Android App:** Standalone cross-platform distribution.

## Industry Collaboration

The project receives industry-level architecture mentorship and technology guidance through **ÉLROI Automation Pvt Ltd**, founded by **Dr. R. Annie Uthra**, contributing to:

- Strategic alignment with real-world Tamil LangTech applications and scalable deployment architectures.
- Mentorship on ethical AI, zero-hallucination guardrails, and enterprise-grade system resilience.

## Academic Guidance

The project is guided by distinguished faculty members from the **Department of Computational Intelligence (CINTEL), School of Computing, SRM Institute of Science and Technology (SRMIST)**:

- **Dr. K. Shanmugam**
- **Dr. R. Udendhran**
- **Dr. R. Babu**
- **Dr. G. Dinesh**

Their academic stewardship ensures structured research methodology, algorithmic rigor, linguistic accuracy, and strict adherence to responsible AI ethics.

> *"AI வழியாக திருக்குறளின் உலகளாவிய வாழ்வியல் நெறிகளை இளைய தலைமுறைக்கும் உலகச் சமூகத்திற்கும் கொண்டு செல்லும் சீரிய முயற்சி."*

## Future Scope

- **Voice-Based Interaction:** Integration of Tamil Speech-to-Text (STT) and expressive Text-to-Speech (TTS) for acoustic recitation and vocal query processing.
- **Expansion Across Classical Epics:** Scaling knowledge modules to include *நாலடியார் (Naladiyar)*, *சிலப்பதிகாரம் (Silappathikaram)*, and the complete *Sangam Corpus (பதினெண்கீழ்க்கணக்கு & எட்டுத்தொகை)*.
- **Adaptive Assessment Engine:** Automated competitive exam question generation and interactive Kural quizzes.
- **Digital Education & LMS Integration:** Connectors for DIKSHA portal, government educational systems, and Tamil Virtual Academy.

> *"திருக்குறள் ஞானம் முழுவதும் ஒரே இடத்தில் - Thirukkural AI"*

## Alignment with Tamil Computing Centre (TCC - SRMIST) – Vision & Mission

Thirukkural AI aligns directly with the core objectives of the **Tamil Computing Centre (TCC - SRMIST)**:

- Development and open-source distribution of AI software and toolkits for Tamil computing.
- Advanced research in Tamil NLP, semantic retrieval, and multi-modal understanding.
- Preservation of classical Tamil heritage and empowerment of Tamil speakers worldwide.

### Activities Completed So Far:

- Conducted the **International Conference on Tamil Computing and Information Technology (ICTCIT) 2025**.
- Organized the premier hackathon **Tamizh-A-THON 1.0**.
- Actively engaged in Tamil book digitization and translation initiatives.
- Published technical contributions in the Tamil computing magazine **"கணினிச்சிறகு"**.

## Conclusion

**Thirukkural AI** is not merely an AI product but a cultural preservation and educational empowerment initiative powered by responsible artificial intelligence. By combining authentic Tamil scholarship with modern Retrieval-Augmented Generation, the project democratizes access to Thirukkural while upholding uncompromising academic integrity.

> *"தமிழின் பெருமையை தொழில்நுட்பத்தின் மூலம் பாதுகாக்கும் மற்றும் பரப்பும் ஒரு பொறுப்பான AI முயற்சி."*